

BPHO ROUND 0 · MARKSCHEME

Drill bank section 7

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.*

Quick key

Q1 C	Q2 A	Q3 D	Q4 B	Q5 E
Q6 A	Q7 C	Q8 E	Q9 B	Q10 D
Q11 E	Q12 B	Q13 A	Q14 C	Q15 D
Q16 B	Q17 D	Q18 A	Q19 E	Q20 C
Q21 D	Q22 B	Q23 C	Q24 C	Q25 B

Worked solutions

Q1 · Dimensional analysis by solving for the exponents: the combination of h , c and G that has the dimensions of a length — **Answer C**

What is being tested. Whether you treat dimensional analysis as an algebra rather than as recognition. The combination is not one to be spotted; it is one to be solved for, and with three constants and three base units it is unique.

Step 1 — write down what you are looking for. Suppose the answer is $h^a c^b G^d$ times some number with no units. Three unknowns, a , b and d . Each base unit gives one equation, so three equations for three unknowns: no freedom left.

Step 2 — one equation per base unit. Read the exponents off the dimensions given in the question.

kilogram: $a - d = 0$
 metre: $2a + b + 3d = 1$
 second: $-a - b - 2d = 0$

Step 3 — the kilogram equation removes one unknown. It says $a = d$, so replace every d by a :

metre: $3a + b = 1$
 second: $-3a - b = 0$

Step 4 — solve the pair. Adding the two equations gives $0 = 1$, so they are not independent in the way they look: the second says $b = -3a$, and putting that into the first gives $3a - 3a = 1$.

That cannot be right, so the metre equation has been mis-copied. Written out properly, the metre exponent is

$2a + b + 3d = 1$ with $d = a$, so $5a + b = 1$

and the second equation still gives $b = -3a$. Then $5a - 3a = 1$, so $2a = 1$ and

$$a = 1/2, \quad d = 1/2, \quad b = -3/2$$

So the combination is $h^{1/2} G^{1/2} c^{-3/2}$, which is

$$\sqrt{(hG/c^3)}$$

So **Answer: C**.

Sanity check. Multiply the kilogram exponents: $1/2 - 1/2 = 0$, so the kilogram cancels, as it must for a length. Seconds: $-1/2 - 2(1/2) + 3(3/2) = -1/2 - 1 + 9/2 = 0$. Metres: $2(1/2) + 3(1/2) - 3(3/2) = 1 + 3/2 - 9/2 = -2$, and the square root halves that to -1 . A length, and only a length, survives.

The distractors.

· $\sqrt{(hc/G)}$ multiplies by c where it should divide by a power of it. The metres come out at $2 + 1 - 3 = 0$ and the seconds at $-1 - 1 + 2 = 0$, so the whole thing is dimensionless and the root of it is a pure number.

· $\sqrt{(hG/c)}$ divides by c only once. The seconds come to $-1 - 2 + 1 = -2$, so the root has seconds to the first power -- a speed, not a length.

· $\sqrt{(hc^3/G)}$ puts c cubed on the wrong side of the fraction, so the seconds come out positive and the result has the dimensions of a length cubed per second rather than a length.

· $\sqrt{(hGc)}$ uses each constant once. The kilogram gives $1 - 1 = 0$ and survives, so the root is of a length cubed over a second, which is not a length.

Relevant topics: base dimensions of the fundamental constants; solving for the exponents of a product; why three constants and three base units leave no freedom.

The trap. Guessing a combination and checking it, rather than solving for the exponents. The three equations are the whole question, and the sign of the c exponent is what separates a length from a speed.

Q2 · Order-of-magnitude estimation: the mass of water in a cloud, from its volume and its liquid water content —

Answer A

What is being tested. Whether you can build an estimate out of two or three quantities and then convert the units honestly. The physics is nothing; the arithmetic is a power of ten, and the power of ten is the question.

Step 1 — say what the answer is made of. The mass of water is the volume of the cloud multiplied by the mass of water in each cubic metre:

$$m = V \times (\text{liquid water content})$$

That is the whole idea. Everything after this is arithmetic.

Step 2 — estimate the volume, in metres. A kilometre is 1.0×10^3 m, and the depth is 500 m:

$$V \approx 1.0 \times 10^3 \times 1.0 \times 10^3 \times 5.0 \times 10^2 = 5.0 \times 10^8 \text{ m}^3$$

Note that only the power of ten really matters here. Nobody knows the depth of a cloud to one significant figure; the point is that it is a few hundred metres, not a few kilometres and not a few metres.

Step 3 — multiply by the concentration.

$$m \approx 5.0 \times 10^8 \text{ m}^3 \times 0.50 \text{ g m}^{-3} = 2.5 \times 10^8 \text{ g}$$

Step 4 — convert grams to kilograms. This is the step the question is really about:

$$m = 2.5 \times 10^8 \text{ g} \div 1000 \text{ g kg}^{-1} = 2.5 \times 10^5 \text{ kg}$$

So **Answer: A.**

Sanity check. A quarter of a million kilograms is about 250 tonnes, which is a few tanker lorries of water. That is the right sort of size for a cloud: big enough to rain, small enough that the sky is not falling on us. An answer of $2.5 \times 10^8 \text{ kg}$ would be a million times that, and would mean every cloud contained a small sea.

The distractors.

- $2.5 \times 10^8 \text{ kg}$ is the mass in grams. It is the natural answer to reach if the conversion is skipped, and it is out by a factor of a thousand.
- $2.5 \times 10^2 \text{ kg}$ converts twice, or divides by 10^6 instead of 10^3 . It is a factor of a thousand the other way.
- $2.5 \times 10^6 \text{ kg}$ uses the depth as 5.0 km, multiplying the volume by ten.
- $5.0 \times 10^5 \text{ kg}$ doubles the answer, which happens if the concentration is read as 1.0 g m^{-3} .

Relevant topics: Fermi estimation; the volume of a slab; concentration as mass per unit volume; grams and kilograms.

The trap. Answering in grams. The concentration is given in grams per cubic metre and the options are in kilograms, so the conversion at the end is the whole difference between a cloud and a reservoir.

Q3 · A small fractional error through a cube: the radius of a sphere is measured one per cent too large, so the volume is three per cent too large — **Answer D**

What is being tested. Whether you know that a small fractional change is multiplied by the power when the quantity is raised to a power. The physics is the volume of a sphere, which everyone knows; the skill is the exponent.

Step 1 — write the volume in terms of the radius.

$$V = (4/3)\pi r^3$$

Step 2 — compare the measured volume with the true one. Let the true radius be r and let the measurement be $r(1 + x)$ with $x = 0.010$. Then

$$V_{\text{measured}} / V_{\text{true}} = [r(1 + x)]^3 / r^3 = (1 + x)^3$$

The $4/3$, the pi and the r^3 have all cancelled. That is the reason the question can be answered at all: no absolute length is needed, only the fractional error.

Step 3 — expand for small x . Since x is 0.010, the terms in x^2 and x^3 are a hundred and a thousand times smaller and can be dropped:

$$(1 + x)^3 \approx 1 + 3x = 1 + 0.030$$

Step 4 — read off the answer. The ratio is $1 + 0.030$, so the volume is 3.0% too large.

So **Answer: D**.

Sanity check. Cubing the ratio exactly gives the same three per cent and a further three hundredths of a percentage point on top, so the first-order answer is out by about three parts in ten thousand. The approximation is doing exactly what it should: it is right to the precision the data deserves. Note too that the error is in the same direction as the radius error, because the volume increases with the radius.

The distractors.

- **0.33%** divides by three. That would be right for a *cube root*, and it has the error going the wrong way as well as the wrong size.
- **1.0%** passes the percentage straight through, which is what you would get if the volume were proportional to the radius. It is proportional to the cube of the radius.
- **2.0%** is the error in the *surface area*, which goes as r^2 and so is doubled rather than tripled. It is a real quantity and not the one asked for.
- **9.0%** squares the percentage instead of cubing the ratio. This is the classic slip: 1.0% is 0.010, and cubing 0.010 gives 10^{-6} , not 9%.

Relevant topics: the first-order expansion; fractional errors and powers; why percentages do not simply add.

The trap. Treating the percentages as quantities in their own right. The cube acts on the *ratio* $1 + 0.010$, not on the number 1.0, so the answer is 3×0.010 expressed as 3.0%, and the 9% that comes from squaring 1.0% is the most tempting wrong answer here.

Q4 · A projectile: the ratio of its maximum height to its horizontal range, which depends on the launch angle alone —

Answer B

What is being tested. Whether you can get two standard expressions into a form where they cancel. Neither formula is hard on its own; the question is whether you notice that the $\sin 2\theta$ in the range has to be expanded before the two will divide.

Step 1 — the maximum height. At the top the vertical velocity is zero, so $v^2 = u^2 + 2as$ with $v = 0$ and $a = -g$ gives

$$H = u^2 \sin^2\theta / (2g)$$

Step 2 — the range. The time of flight is twice the time to the top, $2u \sin\theta/g$, and the horizontal velocity is $u \cos\theta$ throughout:

$$R = u \cos\theta \times 2u \sin\theta / g = 2u^2 \sin\theta \cos\theta / g$$

Step 3 — divide. The u^2 and the g are in both expressions, so they cancel; one power of $\sin\theta$ cancels too:

$$\begin{aligned} H/R &= [u^2 \sin^2\theta / (2g)] \div [2u^2 \sin\theta \cos\theta / g] \\ &= \sin^2\theta / (4 \sin\theta \cos\theta) \\ &= \sin\theta / (4 \cos\theta) \\ &= \tan\theta / 4 \end{aligned}$$

So **Answer: B.**

Sanity check. Put $\theta = 45^\circ$. Then $\tan\theta = 1$ and the ratio is 0.25, so the ball rises a quarter as high as it travels. That is the case most people can picture, and it agrees. Now let the angle approach 90 degrees: the ball goes almost straight up, $\tan\theta$ grows without limit, and the range falls to nothing, so the ratio does grow -- which it does. The expression behaves correctly at both ends.

The distractors.

- $\tan\theta / 2$ uses $u^2 \sin\theta \cos\theta / g$ for the range. That is half the range, the distance covered in the time to the top, and it is the most common slip here.
- $\tan\theta$ divides the height by the horizontal distance at the top of the flight, which drops the factor of two twice.
- $2 \tan\theta$ inverts the ratio of the two coefficients, so the division goes the wrong way round.
- $4 \tan\theta$ comes from using $u^2 \sin\theta \cos\theta / 2g$ for the range, which is the half-range with an extra factor of two as well.

Relevant topics: the projectile equations; expanding $\sin 2\theta$; ratios that are independent of the launch speed.

The trap. Dividing by the half-range. The horizontal distance covered by the time the ball reaches the top is $u^2 \sin\theta \cos\theta / g$, which is half the range, and using it gives $\tan\theta/2$ -- an answer that looks tidier than the correct one.

Q5 · A projectile launched horizontally from a height: the tangent of the angle at which it lands — **Answer E**

What is being tested. Whether you can keep the two components of a projectile's velocity separate. The horizontal one never changes; the vertical one is built up by the fall. Students who try to combine them into a single speed lose the angle entirely.

Step 1 — the horizontal component. Gravity acts vertically, so nothing changes the horizontal velocity. At the moment of landing it is still

$$v_{\text{horizontal}} = u$$

Step 2 — the vertical component. The ball is released with no vertical velocity, so it is a simple fall through a height h . Using $v^2 = u^2 + 2as$ with the initial vertical speed zero and $a = g$:

$$v_{\text{vertical}}^2 = 2gh, \quad \text{so} \quad v_{\text{vertical}} = \sqrt{2gh}$$

Notice that u does not appear. However hard the ball is thrown, it takes the same time to fall and arrives with the same vertical speed.

Step 3 — the tangent of the angle. The velocity vector points along the sum of the two components, and the tangent of the angle below the horizontal is the vertical component over the horizontal one:

$$\tan\theta = v_{\text{vertical}} / v_{\text{horizontal}} = \sqrt{(2gh)} / u$$

So **Answer: E.**

Sanity check. Throw it harder and the ball lands shallower: u in the denominator, so a bigger u gives a smaller tangent, which is a smaller angle. Drop it from higher and it lands steeper: h in the numerator. Both are what you would see. And the expression is a pure number, since a speed divided by a speed has no units -- as a tangent must.

The distractors.

- $u / \sqrt{(2gh)}$ is the reciprocal. It is the tangent of the angle *above* the horizontal, which is a real angle in this problem, just not the one asked for.
- $2gh / u$ uses v^2 where v is wanted. The dimensions give it away: $2gh$ has units of $\text{m}^2 \text{s}^{-2}$, so this is not a pure number.
- $\sqrt{(gh)} / u$ uses $v^2 = gh$, which is the formula for a fall of $h/2$.
- $2\sqrt{(gh)} / u$ leaves a factor of two inside the root that belongs outside it, so the vertical component is too big by a factor of $\sqrt{2}$.

Relevant topics: independence of the horizontal and vertical motions; $v^2 = u^2 + 2as$; resolving a velocity into components.

The trap. Trying to find the landing *speed* and then the angle from it, which needs Pythagoras and a square root that never cancels. The tangent is a ratio of the two components, and taking the ratio directly keeps every square root that is needed and none that is not.

Q6 · A perfectly inelastic collision with a stationary target three times as heavy: the fraction of the kinetic energy lost

— **Answer A**

What is being tested. Whether you keep momentum and energy apart. Momentum is always conserved in a collision; kinetic energy is conserved only when the collision is elastic, and the word "stick" rules that out.

Step 1 — momentum, before and after. Before, only the moving particle has momentum. After, the two move together with a common speed v and a combined mass $4m$:

$$\begin{aligned} m u &= (m + 3m) v \\ v &= u/4 \end{aligned}$$

Step 2 — the kinetic energy before.

$$\text{KE}_{\text{before}} = \frac{1}{2} m u^2$$

Step 3 — the kinetic energy after. The mass is four times as big and the speed is four times smaller, so the energy changes by a factor of four from the mass and a factor of sixteen from the speed:

$$KE_{\text{after}} = \frac{1}{2} (4m) (u/4)^2 = \frac{1}{2} (4m) (u^2/16) = m u^2/8$$

Step 4 — the fraction lost. Compare with the energy before, which is $m u^2/2$:

$$KE_{\text{after}} / KE_{\text{before}} = (m u^2/8) \div (m u^2/2) = 1/4$$

A quarter survives, so three quarters is lost:

$$\text{fraction lost} = 1 - 1/4 = 3/4$$

So **Answer: A.**

Sanity check. The answer must be a fraction between zero and one, and $3/4$ is. It must also be larger than a half, because the target is three times as heavy as the projectile and a heavy target destroys a lot of the motion. In the limiting case where the target is infinitely heavy the projectile simply stops and everything is lost; as the target gets lighter and lighter, less is lost. So the answer must lie between the equal-mass case and one, and $3/4$ does.

The distractors.

- **1/4** is the fraction that survives. It comes from the same working, stopped one step early -- and the question asks for what is lost, not what is left.
- **1/2** is the answer when the two masses are equal. It is the case everyone remembers, and it is the wrong case here.
- **2/3** uses the mass ratio the wrong way up in the energy, so that three parts come out lost for every two.
- **3/8** comes from taking the lost energy to be the heavier particle's share of the final kinetic energy, which is $1/8$ of the original, and then mishandling the subtraction.

Relevant topics: conservation of momentum in a perfectly inelastic collision; why kinetic energy is not conserved; the fraction of energy lost as a function of the mass ratio.

The trap. Answering with the fraction that *survives*. The working is identical up to the last line, and $1/4$ is sitting there as an option.

Q7 · A uniform disc with an off-centre circular hole: where its centre of mass sits, treated as a negative mass —

Answer C

What is being tested. Whether you know that a hole can be carried through the arithmetic as a piece of negative mass. It is the standard trick for composite bodies and it turns a two-dimensional integral into one line.

Step 1 — replace the hole by a negative mass. A uniform disc with a hole is the same as a complete disc of the full size *plus* a disc of negative mass where the hole is. The density is the same everywhere, so it cancels and the masses can be replaced by areas.

Step 2 — the areas.

$$\begin{aligned}\text{whole disc: } & \pi(2a)^2 = 4\pi a^2 \\ \text{hole: } & -\pi a^2 \\ \text{remaining: } & 4\pi a^2 - \pi a^2 = 3\pi a^2\end{aligned}$$

Step 3 — take moments about the centre of the disc. Measure x positive towards the hole. The whole disc is centred on the origin, so its contribution is zero. The hole is a distance a along the positive direction, so its contribution is its (negative) area times that distance:

$$4\pi a^2 \times 0 + (-\pi a^2) \times a = 3\pi a^2 \times x$$

Step 4 — divide by the area that is left.

$$\begin{aligned}-\pi a^3 &= 3\pi a^2 x \\ x &= -a/3\end{aligned}$$

The minus sign means the centre of mass has moved the other way, **away from the hole**, by a distance $a/3$.

So **Answer: C**.

Sanity check. The remaining material is not symmetric, and the side with less material must be the side the centre of mass moves away from -- which it does. The distance is $a/3$, which is smaller than a , so the centre of mass is still inside the hole-free part of the disc. And the answer is a length built from the two lengths in the question and nothing else, as it must be.

The distractors.

- **$a/2$, on the same side as the hole** has the direction wrong, which would be right only if material had been *added* there.
- **$a/4$** divides the moment by the area of the whole disc, $4\pi a^2$, instead of by what remains. It is the same working with the wrong denominator, and it gives a smaller shift than the truth.
- **$a/3$, on the same side as the hole** has the size right and the sign wrong: it is the mirror image of the correct answer.
- **$2a/3$** treats the hole as though its area were four times bigger than it is, trebling the moment without changing the remaining area.

Relevant topics: the centre of mass of a composite body; negative mass; moments about an axis; uniform laminae.

The trap. Dividing the moment by the area of the whole disc rather than by the area that is left. The hole is not a small correction here -- it removes a quarter of the material -- so the denominator matters as much as the numerator.

Q8 · Terminal speed of a falling body when the drag is proportional to the square of the speed — **Answer E**

What is being tested. Whether you know what terminal speed *is*. It is not a property of the object; it is the speed at which the drag has grown enough to cancel the weight, so that the net force and therefore the acceleration are zero.

Step 1 — the condition. At terminal speed the body falls at a constant speed, so its acceleration is zero. By Newton's second law the resultant force is zero, and the only two forces are the weight downwards and the drag upwards:

$$\text{drag} = \text{weight}$$

Step 2 — write the balance with the given drag law. The question says the drag is $k v^2$, and the weight is mg :

$$k v^2 = m g$$

Step 3 — solve for the speed. Divide by k and then take the square root:

$$v^2 = m g / k$$

$$v = \sqrt{(m g / k)}$$

So **Answer: E.**

Sanity check. The expression must have the dimensions of a speed. mg is a force, in kg m s^{-2} ; k has units of kg m^{-1} (the units that make $k v^2$ a force); their ratio is $\text{m}^2 \text{s}^{-2}$, and the root of that is m s^{-1} . A speed. It also behaves correctly: a heavier body has a larger terminal speed and a thicker fluid (larger k) has a smaller one.

The distractors.

- mg/k is the answer for a drag proportional to v rather than to v^2 . It is a speed by the same dimensional argument, which is what makes it tempting -- but it does not come from this drag law.
- $k/(mg)$ inverts the ratio. It has the dimensions of an inverse speed, so it cannot be a speed at all.
- $\sqrt{k/(mg)}$ both inverts and takes the root, giving the reciprocal of the correct answer.
- mg/k^2 takes the square root of k alone and leaves it in the denominator, so the units come out wrong.

Relevant topics: terminal speed; Newton's second law with a velocity-dependent force; rearranging a quadratic balance.

The trap. Treating terminal speed as the speed at which the drag *starts*, or as something to be found from the motion rather than from the forces. It is defined by a balance, and the balance is the only equation needed.

Q9 · A cylinder resting in a V-groove with faces at 60 degrees and 30 degrees: the force on the steeper face, which is the smaller of the two — **Answer B**

What is being tested. Whether you resolve along the right directions. A normal reaction acts perpendicular to the surface it comes from, so the angle a face makes with the horizontal is *not* the angle its force makes with the vertical -- it is the complement. Getting that the right way round is the whole question.

Step 1 — the two normals and their angles. The left face rises at 60° to the horizontal, so its normal is at 60° to the *vertical*... read carefully: a normal is perpendicular to the face, so it is at 30° to the vertical and 60° to the horizontal. The right face rises at 30° , so its normal is at 60° to the vertical and 30° to the horizontal. The two normals are at 30° and 60° to the vertical, and they point up and inwards.

Step 2 — resolve horizontally. There is no horizontal acceleration, so the two horizontal components cancel:

$$\begin{aligned} N_{\text{left}} \sin 60^\circ &= N_{\text{right}} \sin 30^\circ \\ N_{\text{left}} (\sqrt{3}/2) &= N_{\text{right}} (1/2) \\ N_{\text{left}} &= N_{\text{right}} / \sqrt{3} \end{aligned}$$

So the left face already carries the smaller force. Everything after this only fixes its size.

Step 3 — resolve vertically. The two vertical components together carry the weight:

$$\begin{aligned} N_{\text{left}} \cos 60^\circ + N_{\text{right}} \cos 30^\circ &= W \\ N_{\text{left}} (1/2) + N_{\text{right}} (\sqrt{3}/2) &= W \end{aligned}$$

Step 4 — substitute and solve. Put $N_{\text{right}} = \sqrt{3} N_{\text{left}}$ from Step 2:

$$\begin{aligned} N_{\text{left}} (1/2) + \sqrt{3} N_{\text{left}} (\sqrt{3}/2) &= W \\ N_{\text{left}} (1/2) + N_{\text{left}} (3/2) &= W \\ 2 N_{\text{left}} &= W \\ N_{\text{left}} &= W/2 \end{aligned}$$

So **Answer: B.**

Sanity check. The right face carries $N_{\text{right}} = \sqrt{3} N_{\text{left}} = W\sqrt{3}/2 \approx 0.87W$. The two add up to $0.50W + 0.87W = 1.37W$, which is more than W -- and that is right, because they are not parallel: only their vertical components add, and those come to W exactly. As a second check, if both faces were at 45° each would carry $W/\sqrt{2} \approx 0.71W$; tilting one face steeper must increase the load on the other, and $0.87 > 0.71$.

The distractors.

- $W/\sqrt{3}$ gives the steeper face the larger share. It is the natural guess -- a steeper wall looks more load-bearing -- and it is backwards, because a steeper face pushes more nearly horizontally and so carries less of the weight.
- $W\sqrt{3}/2$ is the force on the *shallower* face. It is a real quantity in this problem and not the one asked for.
- W splits the weight evenly between the faces, which would be right only if both faces were at the same angle.
- $2W/\sqrt{3}$ resolves the weight along the steeper face instead of perpendicular to it, which is the direction a block would slide in rather than the direction the face pushes.

Relevant topics: normal reactions; resolving a weight into components along two given directions; three forces in equilibrium.

The trap. Reading the angle off the face instead of off the normal. A face at 60 degrees to the horizontal pushes at 30 degrees to the vertical, and taking the two to be the same swaps the two forces -- which is exactly what the first two distractors do.

Q10 • A wheel rolling at constant speed: the acceleration of the point at the top of the rim, which is the same magnitude as at every other point — **Answer D**

What is being tested. Whether you keep a rolling body's *velocity* and its *acceleration* apart. The famous fact about rolling is that the top of the wheel moves at twice the speed of the wheel and the contact point does not move at all. The temptation is to carry that doubling over to the acceleration. It does not apply.

Step 1 — what is the wheel doing? It rolls without slipping at a constant speed. The centre of the wheel therefore moves in a straight line at a steady speed, so **the centre has no acceleration**.

Step 2 — what is a point on the rim doing? In the frame of the wheel, every rim point goes round the centre in a circle of radius r , once per revolution. The angular speed is the same for every point on the rim, because the wheel is rigid. So relative to the centre, every rim point has a centripetal acceleration

$$a = v^2/r \text{ directed towards the centre}$$

Step 3 — put the two together. The acceleration of a rim point in the frame of the road is the acceleration of the centre plus its acceleration relative to the centre. The centre contributes zero, so

$$a_{\text{top}} = v^2/r, \text{ directed straight down towards the centre}$$

So **Answer: D**.

Sanity check. The answer must not depend on which point of the rim you pick, because the question could have asked about any of them and the physics would be the same. v^2/r is indeed the same for all of them; $2v^2/r$ would single out the top, which the physics does not. Dimensional check: v^2/r is $\text{m}^2 \text{s}^{-2}$ over m , which is m s^{-2} , an acceleration.

The distractors.

- **0** is what you get by thinking of the wheel as a body moving at constant speed. A point on the rim is going round a circle, and anything going round a circle is accelerating even at constant speed.
- $2v^2/r$ uses the top point's *speed*, which really is $2v$, in the formula for acceleration. The velocity of the top point is a sum of the translational and rotational parts; the acceleration is not.
- $v^2/(2r)$ uses the diameter where the radius belongs.
- $4v^2/r$ adds a centripetal acceleration to a translational one that does not exist, because the centre is not accelerating.

Relevant topics: rolling without slipping; centripetal acceleration; the difference between the velocity and the acceleration of a point on a rotating body.

The trap. Doubling the acceleration because the top point's speed is doubled. The velocity of a point on a rolling wheel is a sum of two contributions and its acceleration is not; the contact point has zero velocity and the same acceleration as the top.

Q11 • A rod clamped at both ends and heated: the thermal stress, which is the stress needed to undo the expansion that could not happen — **Answer E**

What is being tested. Whether you can see a problem about a rod that cannot move as two imaginary problems about a rod that can. That is the standard way into every thermal-stress question, and it removes the need for any cleverness about forces.

Step 1 — let the rod expand freely. If the clamps were not there, the rod would grow by

$$\Delta L = \alpha L \Delta T$$

Step 2 — push it back. The clamps hold the rod at its original length, so the rod is now compressed by exactly that amount. The strain it carries is

$$\text{strain} = \Delta L / L = \alpha L \Delta T / L = \alpha \Delta T$$

The length has cancelled. That is not a coincidence: a strain is a ratio of two lengths, and the thermal expansion is proportional to the length, so the two always cancel.

Step 3 — turn the strain into a stress. Young's modulus is defined as stress over strain, so

$$\text{stress} = E \times \text{strain} = E\alpha\Delta T$$

So **Answer: E.**

Sanity check. Take a steel rod, $\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$ and $E = 2.0 \times 10^{11} \text{ Pa}$, heated by 50 K. The stress is $2.0 \times 10^{11} \times 1.2 \times 10^{-5} \times 50 = 1.2 \times 10^8 \text{ Pa}$, which is over a hundred megapascals -- a serious stress, and the right sort of size for why railway rails are laid with gaps. Also note that the answer does not mention the length or the cross-sectional area, and neither of those should matter: the stress is the same in a short rod and a long one.

The distractors.

- $E\Delta T/\alpha$ swaps the expansivity and the temperature. It would make a metal with a tiny expansivity develop an enormous stress, which is backwards.
- $\alpha\Delta T/E$ is the strain divided by the modulus, so it has the dimensions of strain over stress rather than stress.
- $E\alpha/\Delta T$ divides by the temperature change, so heating the rod more would *reduce* the stress. It is a real formula in this problem only in the sense that it has the right letters in it.
- $E\alpha\Delta T^2$ squares the temperature change, which comes from using the thermal expansion as though it were a stress rather than a strain.

Relevant topics: linear thermal expansion; strain as a ratio of lengths; Young's modulus; the two-stage method for thermal stress.

The trap. Looking for the force in the rod. The question asks for a stress, and a stress is a strain multiplied by a modulus -- the length and the area never enter, which is why the answer has no geometry in it at all.

Q12 · The speed of sound in a gas compressed at constant temperature: it does not change, because the density rises in step with the pressure — **Answer B**

What is being tested. Whether you look at what a formula contains before deciding what a change does to it. The speed of sound depends on the *ratio* of pressure to density, and a constant-temperature compression changes both by the same factor.

Step 1 — read the formula.

$$v = \sqrt{\gamma P / \rho}$$

The pressure appears, and so does the density. Any question that changes one of them without asking about the other is a trap.

Step 2 — what happens to the density? The gas is compressed at constant temperature, so the volume falls and the pressure rises -- the two go together, which is Boyle's law. The mass is fixed, so if the volume halves the density doubles. The pressure was doubled by the compression, so

$$P \rightarrow 2P \quad \text{and} \quad \rho \rightarrow 2\rho$$

Step 3 — put both changes in.

$$v_{\text{new}} = \sqrt{(\gamma \times 2P / (2\rho))} = \sqrt{(\gamma P / \rho)} = v$$

The factor of two appears once on the top and once on the bottom, and cancels.

So **Answer: B.**

Sanity check. The ideal gas law says $P = \rho RT / m$, so $P / \rho = RT / m$, which depends only on the temperature. That is a much shorter route to the same conclusion, and it explains the result: for a given gas, the speed of sound is fixed by the temperature and by nothing else. It also tells you what would change it -- heat the gas and the speed rises as the square root of the absolute temperature, which is why sound travels faster on a hot day.

The distractors.

- **It doubles.** Reading v as proportional to P ignores the density entirely, and the square root as well.
- **It is halved.** Assumes the density doubles while the pressure does not, which describes a different process.
- **It rises by $\sqrt{2}$.** Takes the square root of the pressure factor and forgets that the density changed too. This is the most defensible wrong answer, because the square root is genuinely there -- it is just operating on a ratio that has not moved.
- **It falls by $\sqrt{2}$.** The same mistake with the density on the wrong side.

Relevant topics: the speed of sound in a gas; the ideal gas law; why pressure and density move together at constant temperature.

The trap. Changing one variable in a formula that contains two. The compression doubles the density as well as the pressure, and an answer that mentions $\sqrt{2}$ has used one of them and dropped the other.

Q13 · Beats between two tuning forks: the direction the beat rate moves when one fork is loaded tells you which side of the known frequency the unknown lies on — **Answer A**

What is being tested. Whether you know that beats give you a difference and not a direction, and whether you can use a second observation to supply the missing sign. Half of the marks are in the second half of the question.

Step 1 — what the beat rate tells you. Four beats per second means the two frequencies differ by 4 Hz. It does not say which is higher, so there are two candidates:

$$f = 512 + 4 = 516 \text{ Hz} \quad \text{or} \quad f = 512 - 4 = 508 \text{ Hz}$$

Step 2 — what waxing does. Adding wax adds mass to the fork without making it stiffer, so it vibrates more slowly. The 512 Hz fork therefore drops to some frequency below 512 Hz.

Step 3 — test the candidate above. Suppose the unknown fork were 516 Hz. The waxed fork is now below 512 Hz, so the gap between 516 and the waxed fork is *larger* than the original 4 Hz. The beat rate would rise. It falls, so 516 Hz is wrong.

Step 4 — test the candidate below. Suppose the unknown fork is 508 Hz. The waxed fork has dropped from 512 Hz towards it, so the gap is now smaller than 4 Hz. It has fallen to 2 Hz, which means the waxed fork sits at

$$508 + 2 = 510 \text{ Hz}$$

That is a drop of 2 Hz from 512, which is exactly what a small amount of wax would do.

So **Answer: A.**

Sanity check. Both measurements are used and neither is contradicted. The first fixes the size of the gap at 4 Hz; the second fixes its direction. Notice also that the answer is consistent with the size of the change: a 2 Hz drop is small, so the two forks started 4 Hz apart and ended 2 Hz apart, which is the right sort of relationship for a modest amount of wax. If the waxed fork had ended at, say, 480 Hz, the beat rate would have been 28 Hz and the question would have said so.

The distractors.

- **516 Hz** is the other candidate from the first measurement. It is the single most likely wrong answer, because it uses the first observation and ignores the second.
- **510 Hz** is the frequency of the *waxed* fork, not of the unknown one. It is a real number in this problem and it answers a different question.
- **514 Hz** adds the new beat rate to the original fork's frequency, mixing a measurement taken before the wax with one taken after.
- **520 Hz** comes from doubling the beat rate or adding both beat rates together.

Relevant topics: beats as a difference in frequency; the effect of loading a tuning fork; using a change of sign to choose between two candidates.

The trap. Stopping at 516 Hz. The first measurement leaves two candidates and the second is what decides between them; an answer that uses only the beat rate has used half the question.

Q14 · The least height of a plane mirror in which a person can see their whole reflection, which is half their height and independent of the distance — **Answer C**

What is being tested. Whether you draw the geometry rather than guess from experience. Almost everyone's instinct is that the mirror must be nearly as tall as you are, and the geometry says it need be only half.

Step 1 — where is the image? A plane mirror puts the image as far behind the mirror as the object is in front of it, and the image is the same size and the right way up. So there is a virtual person standing

behind the glass, mirrored, at the same distance.

Step 2 — trace the two extreme rays. For the person to see the top of their head, a ray must leave the top of the head, reflect off the mirror, and enter the eye. For the feet, the same. Each ray obeys the law of reflection, so each one goes to the eye along a path that is symmetric about the mirror.

Step 3 — find where each ray meets the mirror. For the ray from the top of the head to the eye, the mirror lies halfway between the head and the eye *in the direction perpendicular to the mirror*. That means the reflection point is at a height halfway between the eye and the top of the head. Likewise, the reflection point for the feet is at a height halfway between the eye and the feet.

top of the mirror: $(\text{head height} + \text{eye height})/2$

bottom of the mirror: $(\text{eye height} + 0)/2$

Step 4 — the height of the mirror is the difference.

mirror height = $(\text{head} - \text{eye})/2 + (\text{eye} - 0)/2 = \text{head}/2$

The eye height has cancelled. So the mirror needs to be half the person's height:

$1.70 \text{ m} / 2 = 0.85 \text{ m}$

So **Answer: C.**

Sanity check. The answer has the dimensions of a length, as it must. It does not contain the distance from the mirror, and that is the surprising part -- but it is right: walk backwards and every ray simply lengthens in proportion, so the reflection points do not move. And the mirror's position is determined too: its bottom edge sits halfway between the eye and the floor, and its top edge halfway between the eye and the top of the head. Those two positions depend on where the eye is, but the *gap* between them does not.

The distractors.

- **1.70 m** is the instinctive answer: the mirror as tall as the person. It is what a shop selling full-length mirrors wants you to believe.
- **0.43 m** is a quarter of the height, which comes from halving twice instead of once.
- **0.57 m** is a third of the height, which is what you get if you halve twice -- for instance by treating the eye as being at the top of the head and then halving again.
- **It depends on the distance** is the trap for anyone who reasons from experience instead of from the diagram. The similar triangles show that the distance cancels exactly, so the answer is the same however far away the person stands.

Relevant topics: the law of reflection; the position of an image in a plane mirror; similar triangles in ray diagrams.

The trap. Reasoning from experience rather than from the diagram. The mirror needs to be half your height, its position depends on where your eye is, and the distance you stand at cancels out of the working completely.

Q15 · The smallest angle a telescope objective can resolve, which is set by the ratio of the wavelength to the diameter

— **Answer D**

What is being tested. Whether you can do a division in powers of ten without a calculator. The physics is given to you in the stem; the arithmetic is the question.

Step 1 — write down the ratio.

$$\theta \approx \lambda / D$$

Step 2 — put the two lengths in the same form, digits separate from powers of ten:

$$\lambda = 5.5 \times 10^{-7} \text{ m}$$

$$D = 0.10 \text{ m} = 1.0 \times 10^{-1} \text{ m}$$

Writing the diameter as 1.0×10^{-1} rather than as 0.10 is the whole trick: it makes the exponent explicit so it cannot be lost.

Step 3 — divide the digits and subtract the exponents.

$$\theta = (5.5 / 1.0) \times 10^{-7 - (-1)} = 5.5 \times 10^{-6} \text{ rad}$$

So **Answer: D.**

Sanity check. An angle in radians is a ratio of two lengths, so it should have no units -- and it does not. It should also be small: a radian is about 57 degrees, so 5.5×10^{-6} rad is about 3×10^{-4} degrees, roughly a second of arc. That is the right order for a small amateur telescope, which is reassuring rather than accidental: the wavelength of light is about half a micrometre and the aperture is a tenth of a metre, and their ratio is a few millionths.

The distractors.

- 5.5×10^{-7} rad is the wavelength itself, so the division by the diameter has been left out entirely.
- 1.1×10^{-5} rad uses the radius, 0.050 m, instead of the diameter. The denominator halves and the angle doubles.
- 2.8×10^{-6} rad is exactly half the right answer, which comes from dividing by the diameter and then halving again for no reason.
- 5.5×10^{-5} rad is ten times too big: the exponent has been added rather than subtracted, so the division went the wrong way.

Relevant topics: the diffraction limit of a circular aperture; radians as a ratio of lengths; multiplying and dividing powers of ten.

The trap. Losing the exponent. Writing the diameter as 0.10 rather than as 1.0×10^{-1} makes it easy to divide 5.5 by 10 and come out a factor of ten wrong; the answer is a power of ten either way, which is why the distractors are powers of ten too.

Q16 · The cost of running a heater, built as a chain of rates: power, then hours, then days, then a price per kilowatt-hour — **Answer B**

What is being tested. Whether you can build a chain of rates and keep the units straight. There is no formula to remember; the units tell you what to multiply by what.

Step 1 — what unit does the answer need? The price is 15 p per **kilowatt-hour**. So the energy must come out in kilowatt-hours, and a kilowatt-hour is simply a kilowatt multiplied by an hour -- which is why the power is given in kilowatts and the time in hours.

Step 2 — the energy for one day.

$$2.0 \text{ kW} \times 3.0 \text{ h} = 6.0 \text{ kWh per day}$$

Step 3 — the energy for the month.

$$6.0 \text{ kWh per day} \times 30 \text{ days} \approx 180 \text{ kWh}$$

Step 4 — the cost.

$$180 \text{ kWh} \times 15 \text{ p per kWh} = 2700 \text{ p}$$

and 2700 p is £27.

So **Answer: B.**

Sanity check. £27 a month for one heater is the right sort of figure for a domestic electricity bill, which is the check that matters most here. Notice also that the answer never needed the number of joules in a kilowatt-hour: the price was quoted per kilowatt-hour, so working in kilowatt-hours kept the whole calculation in the units the price was given in. Converting to joules first would have meant converting back, and would have been three and a half million times bigger in the middle for no benefit.

The distractors.

- **£9** is one day's cost times ten, or three days' cost -- some part of the chain has been dropped.
- **£54** doubles the answer, which happens if the daily use is read as 6.0 hours or the price as 30 p.
- **£90** comes from using 100 p to the pound instead of the correct number of pence.
- **£270** forgets to convert from pence to pounds at all, so the answer is really 2700 p read as pounds.

Relevant topics: electrical power and energy; the kilowatt-hour; chaining rates; converting pence to pounds.

The trap. Converting to joules. The price is per kilowatt-hour, so the energy should be found in kilowatt-hours; converting to joules and back is where the powers of ten get lost, and where the answer becomes a number no one can sanity-check.

Q17 · Two identical cells in parallel driving a resistor: the e.m.f. stays at one cell's value while the internal resistance halves — **Answer D**

What is being tested. Whether you know what cells in parallel do. The instinct is to add the electromotive forces, which is what happens in series; in parallel the e.m.f. stays put and the internal resistance falls.

Step 1 — what is the e.m.f. of the pair? Both cells are connected between the same two nodes, the same way round. Each one on its own would hold those nodes 1.5 V apart. Neither can hold them further apart than the other, so the pair holds them at 1.5 V. **The e.m.f. does not add.**

Step 2 — what is the internal resistance of the pair? The two internal resistances sit between the same two nodes, so they are in parallel:

$$r = (1.0 \times 1.0) / (1.0 + 1.0) = 0.50 \, \Omega$$

Two equal resistors in parallel give half of one. This is the physical meaning of "the cells share the current": each supplies half, so each drops half the internal voltage it would have dropped alone.

Step 3 — the whole circuit. The source resistance is in series with the load:

$$R_{\text{total}} = 0.50 + 2.0 = 2.5 \, \Omega$$

Step 4 — Ohm's law.

$$I = 1.5 / 2.5 = 0.60 \, \text{A}$$

So **Answer: D.**

Sanity check. The current must be more than it would be with one cell alone, which is $1.5/3.0 = 0.50 \, \text{A}$, and it is: 0.60 A. It must be less than the short-circuit current of one cell, $1.5/1.0 = 1.5 \, \text{A}$, because the load is still there, and it is. So the answer lies where it should. It has also risen by only 20 per cent, not by a factor of two -- which is the right shape for a change that only affects the internal resistance.

The distractors.

- **1.5 A** is the answer for the two cells in *series*: $3.0 \, \text{V}$ across $2.0 + 1.0 + 1.0 = 4.0 \, \Omega$. It is a real current in this circuit, just from the wrong circuit.
- **0.50 A** adds the internal resistances instead of combining them in parallel, giving $2.0 + 1.0 + 1.0 = 4.0 \, \Omega$ with a 1.5 V e.m.f. It is also the current from a single cell.
- **0.75 A** combines the resistances correctly but doubles the e.m.f. as well, so it takes both of the two things that could change and gets one of them wrong.
- **0.30 A** uses one cell but adds the other cell's resistance, which no connection of two cells produces.

Relevant topics: cells in parallel and in series; internal resistance; Ohm's law for a whole circuit.

The trap. Adding the electromotive forces. Two identical cells in parallel give one cell's e.m.f. and half its internal resistance; only cells in series add their e.m.f.s.

Q18 · A potentiometer used to compare two e.m.f.s: the balance lengths are in the same ratio as the e.m.f.s, so the wire's length and its resistance never enter — **Answer A**

What is being tested. Whether you understand what a balance point measures. It measures the potential difference along a stretch of a uniform wire, and because the wire is uniform, that potential difference is proportional to the length.

Step 1 — why the length is proportional to the e.m.f. A steady current flows through the whole wire, so the potential difference per centimetre is the same everywhere along it. A length L therefore has a

potential difference proportional to L , with the constant of proportionality set by the driving cell and the wire's resistance per metre.

Step 2 — why that measures an e.m.f. and not a terminal p.d. At balance the galvanometer reads zero, so no current is drawn from the cell being tested. With no current there is no voltage lost across its internal resistance, and the potential difference it holds is its full e.m.f. That is the whole point of the instrument.

Step 3 — form the ratio. The same constant of proportionality applies to both measurements, so it cancels:

$$E_{\text{unknown}} / E_{\text{standard}} = L_{\text{unknown}} / L_{\text{standard}}$$

Step 4 — substitute and solve.

$$\begin{aligned} E_{\text{unknown}} &= 1.5 \text{ V} \times 60.0 / 45.0 \\ E_{\text{unknown}} &= 1.5 \times 4/3 = 2.0 \text{ V} \end{aligned}$$

So **Answer: A.**

Sanity check. The unknown's balance point is further along the wire, so its e.m.f. must be larger than 1.5 V, and 2.0 V is. The ratio 60.0/45.0 is 4/3, so the answer is a third larger than the standard cell, which is the right size for a 1.5 V cell measured against a 2.0 V one. Note also that the 1.00 m length of the wire never entered the working: both balance points are well inside it, and the length only matters in that it must be long enough to reach both.

The distractors.

- **1.1 V** uses the *difference* of the two balance lengths, 15.0 cm, as a fraction of something. The difference is a length, and the ratio that matters is of lengths to the same wire.
- **1.8 V** inverts the ratio, using 45.0/60.0, which makes the unknown smaller than the standard cell even though its balance point is further along.
- **2.5 V** uses the whole metre in place of the standard cell's balance length.
- **3.0 V** adds the two balance lengths, which puts the unknown's balance point beyond the end of the wire.

Relevant topics: the potentiometer; balance and the null method; proportionality along a uniform wire.

The trap. Using the balance lengths as lengths rather than as a ratio. The absolute lengths are irrelevant -- only their ratio is -- and an answer that has used 1.00 m or the difference of the two has used a length where a fraction of the wire was wanted.

Q19 · Charging a capacitor from a battery through a resistor: half of the energy the battery supplies ends up in the resistor, whatever the resistance is — **Answer E**

What is being tested. Whether you can account for energy honestly in a circuit that is not in a steady state. The battery supplies one amount, the capacitor stores another, and the difference has to be somewhere -- it is in the resistor, as heat.

Step 1 — the charge that moves. When the capacitor is fully charged it holds

$$Q = C V$$

That charge has passed through the battery, because the circuit is a single loop.

Step 2 — the energy the battery supplies. The battery lifts every coulomb through its full e.m.f. of V , from the moment the first one moves to the moment the last one does. So

$$\text{energy supplied} = Q V = C V^2$$

Step 3 — the energy the capacitor stores.

$$\text{energy stored} = \frac{1}{2} C V^2$$

The factor of a half is there because the potential difference across the capacitor *rises* from zero to V as it charges, so on average each coulomb is lifted through only $V/2$.

Step 4 — the difference. Whatever is not stored must have been dissipated in the resistor:

$$\text{energy dissipated} = C V^2 - \frac{1}{2} C V^2 = \frac{1}{2} C V^2$$

As a fraction of what was supplied:

$$(\frac{1}{2} C V^2) / (C V^2) = 1/2$$

So **Answer: E.**

Sanity check. The resistance does not appear anywhere in the working, and that is the surprising and correct result: R sets how long the charging takes, not how much energy is lost. A large R means a small current for a long time and the same total heat; a small R means a large current briefly. Doubling the battery's e.m.f. quadruples both the energy supplied and the energy lost, so the fraction stays at a half. The answer is a pure number, as a fraction must be.

The distractors.

- **1/4** is a quarter, which is what you get by squaring the half somewhere -- for instance by comparing the stored energy with the square of the supplied energy.
- **3/4** inverts the question and reports the fraction that *reaches the capacitor*, not the fraction lost.
- **1** assumes everything supplied is dissipated, which would leave the capacitor uncharged.
- **It depends on R** is the trap for anyone who thinks the answer should mention every quantity in the question. It does not: the resistance is a rate-setter, not an energy-taker.

Relevant topics: energy stored in a capacitor; the energy supplied by a battery; conservation of energy in a circuit that is charging.

The trap. Looking for the resistance. The answer is one half and the resistance is not in it, because R sets how long the charging takes rather than how much energy is lost -- and an answer that mentions R has answered a question about time.

Q20 · A lead ball dropped from a height: the rise in temperature, in which the mass cancels and only the height and the specific heat capacity matter — **Answer C**

What is being tested. Whether you can set up an energy balance and then notice that one of the quantities cancels. The cancellation is the physics: it is why the answer does not depend on the mass of the ball.

Step 1 — the energy that arrives. The ball falls through a height h , losing gravitational potential energy

$$E = m g h$$

Step 2 — the energy that raises the temperature. Heating a mass m of a material of specific heat capacity c by ΔT needs

$$E = m c \Delta T$$

Step 3 — equate and cancel. If all of the first becomes the second:

$$\begin{aligned} m g h &= m c \Delta T \\ \Delta T &= g h / c \end{aligned}$$

The mass has gone. That is not a mathematical convenience; it says that a heavy ball and a light one dropped from the same height warm up by the same amount, because the heavier one both has more energy and needs more energy for the same rise.

Step 4 — put the numbers in.

$$\Delta T \approx (10 \times 20) / 130 = 200/130 \approx 1.5 \text{ K}$$

So **Answer: C.**

Sanity check. The answer is a temperature, and the units confirm it: gh/c is $(\text{m s}^{-2})(\text{m}) / (\text{J kg}^{-1} \text{K}^{-1})$, and a joule per kilogram is $\text{m}^2 \text{s}^{-2}$, so the whole thing is K. The size is also right: a rise of a degree and a half is small enough that you would not notice it by hand, which matches experience -- dropped objects do warm up, but only slightly. Lead warms more than most metals because its specific heat capacity is low, and a material with a specific heat capacity ten times larger would warm by a tenth as much.

The distractors.

- **0.15 K** divides by the height where it should multiply, so the answer is the reciprocal of the right one.
- **1.0 K** rounds the specific heat capacity to 200, or divides by the height as well as by the capacity.
- **2.0 K** multiplies by g twice, or uses a height of 200 m.
- **15 K** is ten times too big, which happens if the energy per kilogram is read as a temperature without dividing by the specific heat capacity.

Relevant topics: gravitational potential energy; specific heat capacity; energy balances in which the mass cancels.

The trap. Hunting for the mass. It is not given and it is not needed, because it cancels -- and an answer that looks for it has missed the reason the question can be asked at all.

Q21 • A sealed tyre warmed by a journey: the gauge pressure rises by a larger fraction than the temperature, because the gauge shows only the excess over the atmosphere — **Answer D**

What is being tested. Whether you keep the two pressures apart. A tyre gauge does not measure the pressure of the air inside; it measures how much more there is than the air outside. The gas law applies to the first and the gauge shows the second, and the whole question lives in that gap.

Step 1 — what is held fixed. The tyre does not change shape, so the volume of the air inside is constant. For a fixed mass of gas at constant volume the pressure is proportional to the absolute temperature:

$$p / T = \text{constant}$$

Step 2 — put the temperatures on the absolute scale. A ratio of temperatures is only meaningful in kelvin, because the Celsius scale starts somewhere arbitrary:

$$\begin{aligned} T_1 &= 20 + 273 = 293 \text{ K} \\ T_2 &= 50 + 273 = 323 \text{ K} \end{aligned}$$

Step 3 — the gauge is not the pressure. The gauge reads the excess over the atmosphere, so the pressure that the gas law applies to is

$$p_1 = 2.0 + 1.0 = 3.0 \text{ atm (absolute)}$$

This is the step the question is built on. The gauge says 2.0, but the air inside is pushing with 3.0, and it is the 3.0 that grows with temperature.

Step 4 — apply the law.

$$p_2 / p_1 = T_2 / T_1, \quad \text{so} \quad p_2 = 3.0 \times 323/293$$

Step 5 — evaluate the ratio. The temperature has risen by 30 K on a base of 293 K, so the ratio is one plus a small fraction:

$$323/293 = 1 + 30/293 \approx 1 + 0.10 = 1.10$$

That is the small-change approximation doing its usual work: a rise of about a tenth in absolute temperature gives a rise of about a tenth in absolute pressure.

$$p_2 \approx 3.0 \times 1.10 = 3.3 \text{ atm (absolute)}$$

Step 6 — back to the gauge. The gauge still reads the excess over the atmosphere, so subtract it again:

$$p_2(\text{gauge}) = 3.3 - 1.0 = 2.3 \text{ atm}$$

So Answer: D.

Sanity check. The absolute pressure rose from 3.0 to 3.3, which is 10% and matches the 10% rise in absolute temperature. The gauge, though, went from 2.0 to 2.3, which is 15%. The gas obeyed the law exactly; it is the offset that changed the percentage. That is worth noticing, because it is the reason the answer is 2.3 and not 2.2. And the direction is right: a tyre really does gain a few tenths of an atmosphere on a long drive, which is why pressures are specified for cold tyres.

The distractors.

· **1.7 atm** inverts the temperature ratio. Dividing by 1.10 instead of multiplying gives 2.7 absolute, which is 1.7 on the gauge -- a tyre that cools as it warms.

- **2.0 atm** reads "the volume is constant" as "the pressure is constant". A fixed volume is exactly the condition under which the pressure *must* rise.
- **2.2 atm** scales the gauge reading alone: $2.0 \times 1.10 = 2.2$. It is the answer to a question in which the atmosphere does not exist, and it is the most tempting wrong answer here.
- **3.3 atm** is the absolute pressure. The working is right and the last step is missing, which is why it is a plausible option rather than an absurd one.

Relevant topics: the gas laws at constant volume; the kelvin scale; gauge and absolute pressure.

The trap. Doing the gas law on the gauge reading. The gauge measures an excess, not a total, so a 10% rise in the gas's own pressure becomes a 15% rise on the dial -- and the answer that skips the offset, 2.2 atm, is the one that looks tidiest.

Q22 · A decay curve recorded with a steady background: the background must be subtracted before the curve can be read, and only then does it halve — **Answer B**

What is being tested. Whether you can read a graph for what it is measuring rather than for what it looks like. The detector records the source *and* the room, and the room does not decay.

Step 1 — say what the detector is counting. Every count that arrives during the measurement is either from the source or from somewhere else:

$$\text{recorded rate} = \text{source rate} + \text{background}$$

The background is 200 counts per minute and it is steady. The source is the only thing that changes.

Step 2 — take the background off. Both readings have to come off the curve first, and a curve is read to the nearest gridline, so each of them is an estimate as well as a reading: roughly 840 at the start and roughly 240 at the end. Rearranging, and putting those two in:

$$\begin{aligned} \text{source rate} &= \text{recorded rate} - 200 \\ \text{at } t = 0: \quad 640 &\approx 840 - 200 \\ \text{at } t = 80 \text{ s: } 40 &\approx 240 - 200 \end{aligned}$$

Those are the numbers the curve was hiding. The recorded curve fell from 840 to 240 -- a factor of only 3.5, which is not a power of two at all and is why the curve looks wrong if it is read as drawn.

Step 3 — how many halvings?

$$640 / 40 = 16 = 2^4$$

Four factors of two, so four half-lives have passed in the 80 seconds the source was observed.

Step 4 — one half-life.

$$\text{half-life} = 80 \text{ s} / 4 = 20 \text{ s}$$

So **Answer: B.**

Step 5 — check it against a point in the middle. This is worth doing, because two points can be joined by any number of curves and a third settles it. At $t = 20 \text{ s}$, one half-life in, the source should be at 320

counts per minute. Adding the background back gives $320 + 200 = 520$, and the graph shows 520 at 20 s. The reading is consistent. At $t = 40 \text{ s}$ the source should be at 160, so the recorded rate should be 360, which is also what the graph shows.

Sanity check. Notice what the background does to the answer. If the curve were read as drawn, 840 to 240 is a factor of 3.5, which is between two and three half-lives, so the half-life would come out somewhere near 30 s -- and it would depend on which pair of points you chose, which no half-life ever does. That dependence on the choice of points is the signature of an uncorrected background, and it is the reason a practical physicist measures the background first.

The distractors.

- **10 s** halves the right answer, which is what happens if each 20-second interval is taken as two halvings rather than one.
- **30 s** ignores the background and looks for the point where the recorded curve is half of 840. That is 420, reached at about 30 s, and it is the most common error here.
- **40 s** reads the quarter point of the recorded curve, which the background pushes out past where it belongs.
- **80 s** takes the whole plotted range as one half-life, which reads the graph as a straight line rather than as a decay.

Relevant topics: radioactive decay and half-life; background count rate; reading a decay curve; recognising a ratio as a power of two.

The trap. Reading the curve as drawn. A steady background has been added to something that halves, and the sum does not halve -- it flattens. The tell is that the tail stops falling and the apparent half-life grows as the source weakens.

Q23 · The energy released by one fission, estimated from the rise in binding energy per nucleon rather than from a remembered figure — **Answer C**

What is being tested. Whether you know where the energy of a fission comes from, rather than remembering a number. The source is the binding energy, and the size of the release is set by how much that binding energy rises.

Step 1 — say where the energy comes from. The fragments are more tightly bound than the nucleus they came from. A nucleon that is more tightly bound has less energy, and the difference has to go somewhere: it becomes the kinetic energy of the fragments. So

$$\text{energy released} = \text{increase in total binding energy}$$

Step 2 — turn per-nucleon figures into totals. The binding energy per nucleon is an average over the nucleus, so the total is that figure multiplied by the number of nucleons:

$$\text{total binding energy} = (\text{binding energy per nucleon}) \times A$$

Step 3 — the change per nucleon.

$$\Delta(\text{binding energy per nucleon}) = 8.5 - 7.6 = 0.9 \text{ MeV}$$

Notice how small it is. Less than one MeV per nucleon, against the eight or so that each nucleon already has.

Step 4 — multiply by every nucleon. All 236 of them are in the fragments, and each one gains the same 0.9 MeV:

$$E \approx 0.9 \text{ MeV} \times 236 \approx 212 \text{ MeV}$$

So **Answer: C**.

Sanity check. Two hundred million electronvolts is the standard order of magnitude for a fission, so the estimate has landed where it should. It is also worth seeing how large that is in everyday units: 210 MeV is about $3.4 \times 10^{-11} \text{ J}$, and a chemical reaction releases a few electronvolts per molecule, so a fission releases something like a million times more energy per event than a chemical reaction does. That factor of a million is why a kilogram of uranium can do what a trainload of coal does. Note too that the answer scales with the mass number, so the estimate does not depend on knowing which nucleus this is.

The distractors.

- **2.1 MeV** multiplies the change per nucleon by two, as though only the two fragments were involved rather than every nucleon in them.
- **21 MeV** loses a factor of ten in the subtraction, reading the change as 0.09 MeV per nucleon instead of 0.9.
- **240 MeV** rounds the change per nucleon up to a whole 1.0 MeV, or multiplies by 250 nucleons rather than 236. Both are small slips and both push the answer the same way.
- **420 MeV** doubles the answer, which is what happens if the two fragments are each credited with the full release.

Relevant topics: binding energy and mass defect; why fission releases energy; estimation from a per-nucleon figure.

The trap. Multiplying by the wrong number of nucleons. The change is per nucleon and there are 236 of them, so the release is a couple of hundred MeV and not a couple -- and the smallness of the per-nucleon change is exactly what makes that multiplication necessary.

Q24 · The photoelectric effect across three steps: the threshold frequency read from a stopping-potential graph gives the work function, and the work function then gives the kinetic energy for a stated wavelength — **Answer C**

What is being tested. Whether you can take one number out of a graph and then leave the graph behind. The question has three separate pieces of physics in it -- a graph to read, a work function to find, and a photon energy to calculate -- and they are joined by a single subtraction.

Step 1 — what the height of the graph measures. The stopping potential is the potential difference that brings the fastest emitted electrons to rest. The work done on one of them is $e V_s$, so the height of the graph is the maximum kinetic energy:

$$KE_{\text{max}} = e V_s$$

Step 2 — why a graph of volts is a graph of energy. The electronvolt is defined so that an electron accelerated through one volt gains one electronvolt. That is this situation in reverse: the electron climbs

back through V_s volts and gives up all of its kinetic energy doing so. So the number on the axis can be read as an energy in electronvolts directly, with no conversion at this stage.

Step 3 — where the graph crosses the axis. At the crossing, $V_s = 0$ and so $KE_{\max} = 0$. The electrons come off with nothing to spare, and that frequency is the threshold. It is the only thing this question takes from the graph.

Step 4 — read the crossing off it. It falls between the 4.0 and 6.0 gridlines, and halfway between them is

$$f_0 = 5.0 \times 10^{14} \text{ Hz}$$

Halfway is not a coincidence to be guessed at. The line is straight, so its crossing can be interpolated from any two points on it, and the two gridlines bracket the crossing evenly.

Step 5 — the work function. At the threshold a photon has exactly enough energy to free the electron and none left over, so the whole of its energy is the work function:

$$\phi = h f_0 = 6.6 \times 10^{-34} \times 5.0 \times 10^{14} = 3.3 \times 10^{-19} \text{ J}$$

Step 6 — keep both energies in joules for now. The answer is wanted in electronvolts, but the subtraction is easier in joules and the conversion can be done once at the end. Converting twice means rounding twice, and two roundings then have to be added. So note the work function and carry on:

$$\phi = 3.3 \times 10^{-19} \text{ J}$$

Step 7 — the energy of the 330 nm photon. This comes from the stem and not from the graph:

$$E = h c / \lambda = (6.6 \times 10^{-34} \times 3.0 \times 10^8) / 3.3 \times 10^{-7} \\ = 6.0 \times 10^{-19} \text{ J}$$

Step 8 — subtract, then convert once. Whatever the photon had beyond the work function becomes kinetic energy:

$$KE_{\max} = 6.0 \times 10^{-19} - 3.3 \times 10^{-19} = 2.7 \times 10^{-19} \text{ J}$$

Dividing by the charge on an electron turns that into electronvolts, and this is a division a candidate can do in their head: twenty-seven over sixteen is one and seven tenths to two figures, so

$$KE_{\max} \approx 1.7 \text{ eV}$$

So **Answer: C.**

Step 9 — check the answer three ways. First, it is positive, so electrons are indeed emitted. Second, it is smaller than the photon energy, as it must be, because the electron had to pay the work function on the way out. Third, the threshold wavelength should be $c/f_0 = 3.0 \times 10^8 / 5.0 \times 10^{14} = 600 \text{ nm}$, and the light used is 330 nm, comfortably shorter, so there is plenty of energy to spare -- which matches a kinetic energy that is a good fraction of the photon energy. Finally, the slope of the graph should be h/e : the line rises a little over two volts across five units of 10^{14} Hz , which is $4.1 \times 10^{-15} \text{ V s}$, and multiplying by the elementary charge gives $6.6 \times 10^{-34} \text{ J s}$, the Planck constant the question supplied. The graph is consistent with the constants, so the reading of the crossing can be trusted.

The distractors.

- **0 eV** assumes the light is at or below the threshold. It is not: the threshold wavelength is 600 nm and the light is 330 nm.
- **1.3 eV** reads the crossing as 6.0×10^{14} Hz instead of 5.0. That makes the work function larger by a fifth, and it is exactly the error the interpolated axis is there to catch.
- **2.1 eV** is the work function itself -- the price of escaping, rather than what is left after paying it.
- **3.8 eV** is the energy of the photon in electronvolts. Reporting it means the electron escaped without paying anything, which would make the work function zero and the graph cross the axis at the origin.

Relevant topics: the photoelectric equation; stopping potential and threshold frequency; the work function; photon energy from wavelength; converting between joules and electronvolts.

The trap. Stopping at the work function. The graph gives the threshold, the threshold gives the work function, and the work function is not the answer -- it is one of the two numbers that have to be subtracted. The other one comes from a wavelength the graph never mentions.

Q25 · Counting photons: the energy of one, the rate a lamp emits them at, and the fraction of a sphere that a small detector covers at a stated distance — **Answer B**

What is being tested. Whether you can build a photon count out of three separate physical ideas and keep the units straight while you do it. There is no single formula here. Each of the three steps is its own question, and the answer is the product.

Step 1 — the power that actually leaves as light. The lamp is rated at 100 W but only a twentieth of that is radiated as light of the stated wavelength; the rest is heat:

$$P = 0.050 \times 100 = 5.0 \text{ W}$$

Step 2 — the energy of one photon. From the wavelength:

$$E = h c / \lambda = (6.6 \times 10^{-34} \times 3.0 \times 10^8) / 5.5 \times 10^{-7} \\ \approx 3.6 \times 10^{-19} \text{ J}$$

That is a little over two electronvolts, which is the right size for a visible photon.

Step 3 — the photons leaving the lamp each second. Power is energy per second, so dividing by the energy of one photon gives photons per second:

$$N = P / E = 5.0 / 3.6 \times 10^{-19} \approx 1.4 \times 10^{19} \text{ s}^{-1}$$

Fourteen million million million photons a second, which is the sort of number that explains why light looks continuous.

Step 4 — how the geometry enters. The lamp radiates equally in all directions, so by the time the photons have travelled 3.0 m they are spread evenly over the surface of a sphere. The detector is a small patch of that surface, so the share of the photons it receives is the share of the sphere's area that it covers -- a ratio of two areas and nothing else. The distance is never compared with the size of the detector, which is what makes the step safe to estimate.

Step 5 — the area of that sphere.

$$A_{\text{sphere}} = 4 \pi r^2 = 4 \pi \times (3.0)^2 \approx 113 \text{ m}^2$$

Step 6 — the detector's area, in the same units. This is the step that decides the answer:

$$1 \text{ cm} = 1 \times 10^{-2} \text{ m}, \text{ so } 1 \text{ cm}^2 = 1 \times 10^{-4} \text{ m}^2$$

$$A_{\text{detector}} = 2.0 \times 10^{-4} \text{ m}^2$$

The conversion is squared because the area is a length squared. Two square centimetres is not a hundredth of a square metre; it is two ten-thousandths of one.

Step 7 — the fraction the detector catches.

$$\text{fraction} = A_{\text{detector}} / A_{\text{sphere}} = (2.0 \times 10^{-4}) / 113 \approx 1.8 \times 10^{-6}$$

Step 8 — multiply.

$$\text{photons on the detector} = 1.4 \times 10^{19} \times 1.8 \times 10^{-6}$$

$$\approx 2.5 \times 10^{13} \text{ per second}$$

So Answer: B.

Step 9 — check it. The answer should be far smaller than the lamp's output, and it is: twenty-five million million against fourteen million million million is a factor of about six hundred thousand, and the geometry accounts for it, since the sphere is about half a million times the detector's area. The units work: watts divided by joules gives per second, and an area divided by an area is a pure number, so the result is a rate of photons and nothing else. Twenty-five million million photons a second is a large number, but a detector that size is a tiny target -- and it is still enough for a photoelectric current to be measured, which is worth remembering when a question asks whether a photon can be detected at all.

The distractors.

- 2.5×10^9 converts the detector's area twice, using $2.0 \times 10^{-8} \text{ m}^2$. Squaring a conversion is easy to do twice by accident.
- 5.0×10^{14} uses the full 100 W, forgetting that only 5.0% leaves as light of this wavelength.
- 2.5×10^{15} converts centimetres to metres but does not square the conversion, so the area is a hundred times too big.
- 2.5×10^{17} leaves the area in square centimetres altogether, which is ten thousand times too large.

Relevant topics: the energy of a photon; power as energy per second; the inverse-square spreading of light from a point source; converting square centimetres to square metres.

The trap. Treating the detector's area as a number rather than as a length squared. Two square centimetres is $2.0 \times 10^{-4} \text{ m}^2$, and every option a hundred or ten thousand times too large comes from getting that conversion wrong.